Stochastic Processes

Dr. Louis Asiedu

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Chapter 1

Introduction

Mathematical models can be categorized as being probabilistic or deterministic.Among situations where probabilistic models are more suitable, very often a better representation is given by considering a collection or a family of random variables that are indexed by a parameter such as time and space are known as **stochastic processes.**(a random or chance process).

Suppose that we have some system which moves from state to state over "time". We define the random variables X_t or X(t) as the state of the process at time t.

Definition 1.0.0.1. *A stochastic process is a set of random variables $X(t), t \in T$, where T is called the parameter space of the process.*

NB: t is called an indexing parameter.The values assumed by the process are known as States.

Examples:

1. Consumer preferences on monthly basis

   SS= MTN,TIGO,AIRTEL

   TS= January, February,...,December

2. Tossing a coin sequentially ; $X_n, n = 1, 2...$

SS= H,T

TS= 1,2,3,...

3. time series data $X(t), t \in [0, \infty)$ or $X_t : t = 0, 1, ..$

   The above introduction shows that there are four types of stochastic processes.

   (a) Discrete parameter and state spaces

       e.g: Consumer preferences observed on a monthly basis.

   (b) Continuous parameter space and discrete state space

       e.g: Number of students waiting for the bus at any time of the day.

   (c) Discrete parameter space and continuous state space

       e.g: Waiting time of the $n - th$ student arriving at the bus stop.

   (d) Continuous parameter and state spaces

       e.g: Content of a dam over an interval of time

Chapter 2

Probability Distributions

Let $(t_1, t_2, ..., t_n)$ with $t_1 < t_2 < t_3 < ... < t_n$ be a discrete set of points within T. The joint distribution for the process $X(t)$ at these points can be defined as:

$$P(X(t_1) \leq x_1, X(t_2) \leq x_2, .., X(t_n) \leq x_n)$$

.Let t_0 and t_1 be two points in T such that $t_0 \leq t_1$; then we may define the conditional transition distribution function as:

$$F(x_0, x_1, t_0, t_1) = P[X(t_1) \leq x_1 / X(t_0) = x_0]...*$$

When a stochastic process has discrete parameter and state spaces, we may define the transition probabilities as:

$$P_{(i,j)}^{(m,n)} = P(X_n = j / X_m = i)...m \leq n.$$

where i and j are the state spaces. The probabilities are called **transition probabilities**.

Definition 2.0.0.1. *A stochastic process $X(t), t \in T$ is said to be tiime-homogeneous or parameter-homogeneous if the transition distribution function * depends only on the difference $t_1 - t_0$ instead of t_1 and t_0.*

Then we have:

$$F(x_0, x; t_0, t_0 + t) = F(x_0, x, 0, t) \ldots **$$

for $t_0 \in$ T. For convenience, we can write (**) as $F(X_0, X; t)$.The corresponding expression for the process $X_n, n = 0, 1, 2, \ldots$ would then be $P^n_{i,j}$ (you are at i moving to j in n steps)

$$\begin{aligned} P_{i,j}^{(t,t+n)} &= P(X_{n+t} = j/X_t = i) && (2.1)\\ &= P(X_n = j/X_0 = i), \text{when t} = 0 && (2.2)\\ &= P_{i,j}^{(0,n)} && (2.3)\\ &= P_{i,j}^{(n)} && (2.4) \end{aligned}$$

2.1 Markov Processes and Renewal Processes

Consider a finite (or countably infinite) set of $(t_0, t_1, ..., t_n, t)$, $t_0 < t_1 < t_2 < ... < t_n < t$ and $t, t_r \in T$ $(r = 0, 1, 2, .., n)$ where T is the parameter space of the process $X(t), t \in T$ is called **Markov-dependence** if the conditional distribution of $X(t)$ for given values $X(t_1), X(t_2), ..., X(t_n)$ depends only on $X(t_n)$ which is the most recent known values of the process; i.e if:

$$\begin{aligned} P[X(t) \leq x/X(t_n) = x_n, X(t_{n-1}) = x_{n-1}, \cdots, X(t_0) = x_0] &= P[X(t) \leq x/X(t_n) = x_n] && (2.5)\\ &= F(x_n, x; t_n, t) && (2.6) \end{aligned}$$

The stochastic

$$P[X(t) \leq x/X(t_n) = x_n, X(t_{n-1}) = x_{n-1}, \cdots, X(t_0) = x_0] = P[X(t) \leq x/X(t_n) = x_n] \quad *$$

for $t_0 < t_1 < t_2 < \cdots < t$.

A stochastic process satisfying * is called a **Markov process**.

A stochastic process with discrete state and parameter spaces, * becomes:

$$P[X_n = j/X_{n1} = i_1, X_{n2} = i_2, \cdots, X_{nk} = i_k] = P[X_n = j/X_{n1} = i_1]$$

$\forall n > n_1 > n_2 > ... > n_k$ and $n, n_1, n_2, ..., n_k \in T and\ i, j \in S$.

For a Markov process, therefore if the state is known for a specific value of the time parameter t, that information is sufficient to predict the behavior of the process beyond that point. As a consequence of this, we have the following relation:

$$F(x_0, x; t_0, t) = \int_{y \in S} F(y, x, T, t) dF(x_0, y; t_0, T),$$

where $t_0 < T < t$

2.2 The Chapman-Kolmogorov equations

1. For a Markov chain with discrete parameter and state spaces,

$$P_{ij}^{(m,n)} = \sum_{k \in S} P_{ik}^{(m,r)} P_{k,j}^{(r,n)}$$

   where $m < r < n, \forall i, j \in S$. (the forward equation).

2. For a Markov process with aa discrete state space and a continuous parameter space T,

$$P_{ij}(t+s) = \sum_{k \in S} P_{ik}(t) P_{kj}(s)$$

   $\forall s \geq 0, t \geq 0$, where,

$$P_{ij}(t+s) = P(X_{t+s} = j/X_0 = i)$$

Proof:

a

$$
\begin{aligned}
P_{ij}^{(m,n)} &= P(X_n = j/X_m = i) && (2.7)\\
&= \sum_{k\in S} P(X_n = j, X_r = k/X_m = i) \quad \text{by the total probability rule.} && (2.8)\\
&= \sum_{k\in S} P[X_n = j/X_r = k, X_m = i]P[X_r = k/X_m = i] && (2.9)\\
&= \sum_{k\in S} P[X_n = j/X_r = k]P[X_r = k/X_n = i] \quad \text{by the Markov dependency property} && (2.10)\\
&= \sum_{k\in S} P_{k,j}^{(r,n)} P_{i,k}^{(m,r)} && (2.11)\\
&= \sum_{k\in S} P_{i,k}^{(m,r)} P_{k,j}^{(r,n)} && (2.12)\\
& && (2.13)
\end{aligned}
$$

b

$$
\begin{aligned}
P_{ij}^{(t+s)} &= P(X_{t+s} = j/X_0 = i) && (2.14)\\
&= \sum_{k\in S} P(X_{t+s} = j, X_t = k/X_0 = i) \quad \text{by the total probability rule.} && (2.15)\\
&= \sum_{k\in S} P[X_{t+s} = j/X_t = k, X_0 = i]P[X_t = k/X_0 = i] && (2.16)\\
&= \sum_{k\in S} P[X_{t+s} = j/X_t = k]P[X_t = k/X_0 = i] \quad \text{by Markov dependency} && (2.17)\\
&= \sum_{k\in S} P_{k,j}^{(t,t+s)} P_{i,k}^{(0,t)} && (2.18)\\
&= \sum_{k\in S} P_{i,k}^{(0,t)} P_{k,j}^{(t,t+s)} && (2.19)\\
& && (2.20)
\end{aligned}
$$

2.2.1 Some Conventions in Markov Chains

1. First order dependency axiom (Markovian Postulate)
   $P(X_{n+1} = j/X_n = i, X_{n-1} = i-1, ..., X_0 = i_0) = P(X_{n+1} = j/X_n = i), \forall i_0, i1, i2, ..., i$ and j and $\forall$ n $\geq$0.

2. Stationary or time homogeneous postulate.
   $P(X_n = j/X_m = i) = P(X_k = j/X_i = i)$
   i.e $P_{i,j}^{(m,n)} = P_{i,j}^{(l,k)}$
   provided n>m, k>l, n≠k, m≠l and $n - m = k - l$, ∀ i, j ∈ S.

2.2.2 More Conventions

1. State space is usually represented with the index set $0, 1, 2, ..., m-1$ or $1, 2, ..., m$; assuming m discrete states. A similar convention holds for a discrete parameter space.

2. $P_{i,j}^{(t)} = P(X_t = j / X_{t-1} = i)$, one step dependency assumption.This is the probability of state j at time t given state i at time t-1.

   From the time-homogeneous assumption, we have

$$P_{i,j}^{(t)} = P_{i,j} \quad \forall t \in T$$

If the equation in (2) does not hold, we have a non-homogeneous first order Markov Chain, otherwise, we have a stationary (time homogeneous) first order Markov Chain.

In the latter case, if $P_j^(t) = P(X_t = j)$ then it can be shown that $P_j^(t) = \sum_{i \in S} P_i(t-1).P_{j,i}$,   j=1,2,..m

$P_{i,j}(t) = P(X_{t+1} = j / X_t = i) = P_{i,j}$ is the first step transition probability.

Now by TPR,

$$P_j(t) = \sum_{i \in S} P_i(t-1).P_{j,i}, \quad j = 1, 2, ...; t = o, 1, 2...$$

with t=o being the initial time and $P_i(t) = P(X_t = i)$.

In matrix form,$\underset{\sim}{P}(t) = \underset{\sim}{P}(0)\underset{\sim}{p}^t$; where $\underset{\sim}{P}^t$ is the matrix $\underset{\sim}{P}$ raised to the power t.

$\underset{\sim}{P}$= $(P_{i,j})$ satisfy the following postulates :

i $0 \leq P_{i,j} \leq 1$, $\forall$ i, j $\in$ S .

ii $\sum_i Pi, j = 1$,   for each i$\in$ S.

A square matrix which satisfies these two properties is said to be a stochastic matrix or a transition matrix.

E.g:

$$\begin{pmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{pmatrix} \text{is a stochastic matrix.}$$

If in addition, $\sum_{i\in S} P_{i,j} = 1$ for each j∈S, then $\underset{\sim}{P}$ is said to be a doubly stochastic matrix.

E.g:

$$\begin{pmatrix} 0.4 & 0.6 \\ 0.6 & 0.4 \end{pmatrix} \text{is a doubly stochastic matrix.}$$

2.3 Some common Stochastic Processes

:

1. **Bernoulli Process:** This is a process with discrete state and parameter space.Denote by S_n the number of successes in n trials, clearly $\{S_n\}$ is a Stochastic process with state space {0,1,2...} and

$$P(S_n = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, 2...n$$

   The process $\{S_n\}$ is called the Bernoulli Process.

2. **Poisson Process**: This is a process with discrete state and continuous parameter space. Consider the events occurring under the following postulates:

   * Events occuring in non-overlapping interval of time are independent of each other.
   * There is a constant λ such that the probabilities of occurrence of events in a small interval of length Δt are given as follows;

     -P[number of events occurring in (t,t+Δt)=0]=1-$\lambda\Delta t$+ 0(Δt)

     -P[one event occurring in (t,t+Δt)]=$\lambda\Delta t$+ 0(Δt)

     and $P(X(t) = k) = e^{-\lambda t}\frac{(\lambda t)^k}{k!}$; k=0,1,2..

     The time interval between consecutives occurrences of events in a Poisson Process are independent random variables indentically distributed with pdf:

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$\therefore$ the Poisson Process is also called the renewal counting process.

3. **Gaussian Process:** This is a process with continuous state and parameter spaces.

4. **Weiner Process:** This is a process with continuous state and parameter spaces.

Consider a Stochastic Process $X(t)$ with the following properties;

1. The process X(t),t≥0 has stationary independent increments. This means that for $t_1, t_2 \in T$ and $t_1 < t_2$, the distribution of $X(t_2) - X(t_1)$ is same as $X(t_2 + h) - X(t_1 + h)$ for any $h > 0$ and for any non-overlapping time interval (t_1, t_2) and (t_3, t_4) with $t_1 < t_2 < t_3 < t_4$.

2. For any given time interval (t_1, t_2), $X(t_2) - X(t_1)$ is normally distributed with mean 0 and variance $\sigma^2(t_2 - t_1)$.

The n-step transition probability matrix

Let $\underset{\sim}{P}$ be the transition probability matrix of a Markov Chain $X_n, n = o, 1, 2...$ as defined earlier.
Let $P_j^(n)$ be the probability that the process is in the state j after n transitions (unconditional probability).

Denote by the row vector $\underset{\sim}{P}^(n)$, the vector of probabilities $\underset{\sim j}{P}^(n)$, j∈S.The n-step transition probability $P_{i,j}^(n)$ and the unconditional probabilities $P_j^(n)$, i,j ∈S are determined as;

Theorem:

$\underset{\sim}{P}^(n)$=$\underset{\sim}{P}^n$ and $\underset{\sim}{P}^(n)$= $\underset{\sim}{P}^(0)\underset{\sim}{P}^n$.

Proof:

From Chapman-Kolmogorov Equation:

$$P_{i,j}^(r + s) = \sum_{k \in S} P_{i,k}^{(r)} P_{k,j}^{(s)}$$

for given r and s.

Let $r = 1$ and $s = 1$,

$$P_{i,j}^{(2)} = \sum_{k \in S} P_{i,k} P_{k,j}$$

Clearly, $P_{i,j}^{(2)}$ is the $(i,j)th$ element of the matrix product:

$$\underset{\sim}{P}.\underset{\sim}{P} = \underset{\sim}{P}^{(2)}$$

Assume $\underset{\sim}{P}^{(r)}$=$P^{(r)}$, r=1,2,...,n-1

Let $r = n-1$, $s = 1$,

$$P_{i,j}^{(n)} = \sum_{k \in S} P_{i,k}^{(n-1)} P_{k,j} \quad \text{is the (i,j)th element of}$$

$$\underset{\sim}{P}^{n-1}.\underset{\sim}{P} = P^n$$

$$\therefore \underset{\sim}{P}^{(n)} = P^n$$

For a two-state chain with state space $S = 0, 1$, then the transition matrix is denoted by:

$$\underset{\sim}{P} = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix}$$

Theorem 2.3.0.1. *For a two state chain with transition matrix*

$$\underset{\sim}{P} = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix}, \underset{\sim}{P}^{(n)} \begin{pmatrix} P_{00}^{(n)} & P_{01}^{(n)} \\ P_{10}^{(n)} & P_{11}^{(n)} \end{pmatrix}$$

Example:

Theorem 2.3.0.2. *For a two-state Markov Chain with one step probability transition matrix,*

$$\underset{\sim}{P} = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, 0 \leq a, b \leq 1, |1-a-b| < 1$$

$$\underset{\sim}{P}^n = \begin{pmatrix} \frac{b}{a+b} + \frac{a(1-a-b)^n}{a+b} & \frac{a}{a+b} - \frac{a(1-a-b)^n}{a+b} \\ \frac{b}{a+b} - \frac{b(1-a-b)^n}{a+b} & \frac{a}{a+b} + \frac{b(1-a-b)^n}{a+b} \end{pmatrix}$$

This theorem can be proven easily by induction.

Definition 2.3.0.3. *The probability vector* $(\pi_0, \pi_1, \pi_2, ..., \pi_n)$ *is said to be a limiting distribution or a steady state or long run distribution of a Markov Chain with transition probability matrix* $\underset{\sim}{P}$ *if*

$$\lim_{n\to\infty} \underset{\sim}{P}^n = \pi \tag{2.21}$$

exists and $(\pi_0, \pi_1, \pi_2, ..., \pi_n)$ *is a row of* $\underset{\sim}{\pi}$ *(where* $\underset{\sim}{\pi}$ *is stable) and:*

$$(\pi_0, \pi_1, \pi_2, ..., \pi_n)\underset{\sim}{P} = (\pi_0, \pi_1, \pi_2, ..., \pi_n)$$

Note: If a Markov Chain has a long-run or steady state distribution, then the long run distribution is also stationary;However a Markov chain can have a stationary distribution without a limiting distribution.

Example: For a two-state Chain with transition matrix $\underset{\sim}{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$\pi.\underset{\sim}{P} = \pi \implies (1/2, 1/2)\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = (1/2, 1/2)$$

$$\pi = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \quad \text{is the stationary distribution}$$

However, $\lim_{n\to\infty} \underset{\sim}{P}^n = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^n = \begin{cases} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \text{if n is odd} \\ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \text{if n is even} \end{cases}$ Hence there exist no long-run limiting distribution.

Questions:

1. Prove that if $\underset{\sim}{P}$ is a k×k stochastic matrix, then $\underset{\sim}{P}^n$ is a stochastic matrix $\forall n = 1, 2, ...$

2. Prove that if $P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$, then

$$\lim_{n\to\infty} \underset{\sim}{P}^n = \pi \tag{2.22}$$

exists and $\pi\underset{\sim}{P}$=$\underset{\sim}{P}\pi$=π, where $0 < a, b < 1 and |1-a-b| < 1$.

3. Prove that if $\underset{\sim}{P}$ and $\underset{\sim}{Q}$ are k×k stochastic matrices, then $\underset{\sim}{P}\underset{\sim}{Q}$ is a stochastic matrix.

4. Prove that if $\underset{\sim}{P}$ is a k×k stochastic matrix, then so is $\underset{\sim}{P}^n, \forall n = 1, 2, ...$

Definition 2.3.0.4. *A Stochastic matrix with identical rows is said to be stable.*

e.g: $\begin{pmatrix} 1/4 & 3/4 \\ 1/4 & 3/4 \end{pmatrix}$ *is a stable stochastic matrix.*

Definition 2.3.0.5. *A vector $(P_1, P_2, ...P_m)$ is said to be a probability vector if $\forall i = 1, 2, ..., m$, $0 \leq P_i \leq 1$ and $\sum_{i=1}^{m} P_i = 1$.*

Definition 2.3.0.6. *A probability vector $(P_0, P_1, P_2, ...P_m)$ is said to be stationary with respect to a stochastic matrix $\underset{\sim}{P}$ if $(P_0, P_1, ...P_m)\underset{\sim}{P}$=$(P_0, P_1, ...P_m)$.*

2.4 Modes of transition and Classification of states.

Definition 2.4.0.1. *State j is said to be accessible from state i if j can be reached from i in a finite number of steps. If two states i and j are accessible to each other, then they are said to communicate.*

In probability terms, these definitions imply:

$i \to j$ (j accessible from i) if for some $n > 0$, $P_{i,j}^{(n)} > 0$

$j \to i$ (i accessible from j) if for some $n > 0$, $P_{j,i}^{(n)} > 0$

$i \leftrightarrow j$ (i and j communicate) if for some $n > 0$, $P_{j,i}^{(n)} > 0$ and for some $m > 0$, $P_{j,i}^{(}m)$.

$i \rightarrow j$ (j not accessible from i) if $\forall n \geq 0, P_{i,j}^{(n)} = 0$

$j \rightarrow i$ (i not accessible from j) if $\forall n \geq 0, P_{j,i}^{(n)} = 0$

$i \longleftrightarrow j$ (i and j do not communicate) if $P_{i,j}^{(n)} = 0$, $\forall n \geq 0$ or $P_{j,i}^{(m)} = 0$, $\forall m \geq 0$

2.4.1 Properties of Communication Relation

1. Reflexivity: $i \longleftrightarrow i$

$$P_{i,j}^{(0)} = \delta_{i,j} = \begin{cases} 1 & , i = j \\ 0 & , i \neq j \end{cases}$$

2. Symmetry: $i \longleftrightarrow j$ then $j \longleftrightarrow i$

3. Transitivity: if $i \longleftrightarrow j$ and $j \longleftrightarrow k$, then $i \longleftrightarrow k$

Proof $\exists r, s \in Z^+ / P_{i,j}^{(r)} > 0 and P_{j,k}^{(s)} > 0$

But $P_{i,k}^{(r+s)} = \sum_{i \in S} P_{i,l}^{(r)} P_{l,k}^{(s)} \geq P_{i,j}^{(r)} P_{j,k}^{(s)} > 0$

Therefore, $i \rightarrow k$.

Reflexivity, Symmetry and transitivity properties together form an equivalence class or equivalence relation.The set of all states of a Markov Chain that communicate(with each other) can therefore be grouped into a single equivalence class.

Markov Chains may have more than one such equivalence class. If there are more than one, then it is not possible to have communicating states in different equivalence classes. However it is possible to have states in one class that are accessible form another class.

Definition 2.4.1.1. *If a Markov Chain has all its states belonging to one equivalence class, it is said to be irreducible.*

Clearly, in an irreducible chain, all states communicate.The period of a state i is defined as the greatest common divisor of all integers $n \geq 1$ for which $P_{i,i}^{(n)} > 0$

Example: If $\underset{\sim}{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ then $\underset{\sim}{P}^n = \begin{cases} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \text{if n is odd} \\ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \text{if n is even} \end{cases}$ State space S=0,1

$P_{0,0}^{(2)} = 1 > 0, P_{0,0}(4) = 1 > 0, P_{0,0}(6) = 1 > 0, ..., P_{0,0}(2k) = 1 > 0, ...k = 1, 2, 3....$ The greatest common divisor of 2,4,6,... is 2. Hence the period of state 0 is 2.(which is the same for state 1).

Note: If two states communicate, then they have the same period. A state is said to be aperiodic if it has period 1.

Theorem 2.4.1.2. *If i and j are states of a Markov Chain and $i \longleftrightarrow j$, then i and j have the same period.*

It follows from the above theorem that, periodicity is also a class property. A class of states with periodicity is said to be aperiodic.

If all the states of a Markov Chain communicates and have period 1, then the chain is said to be irreducible and aperiodic.If $\underset{\sim}{P}$ is a transition matrix of finite Markov Chain which is irreducible and aperiodic, it is easy to show that $\exists n \in Z^+, n \geq 1$, for which $\underset{\sim}{P}^{(n)}$ has no zero element (all states are accessible).The matrix $\underset{\sim}{P}$ is then said to be regular or primitive, the chain also then is said to be <u>regular</u>.

Theorem: Let $\underset{\sim}{P}$ be the transition probability matrix of an irreducible, aperiodic m-state finite time-homogeneous Markov Chain, then:

$$\lim_{n\to\infty} \underset{\sim}{P}^n = \underset{\sim}{\pi} = \begin{pmatrix} \underset{\sim}{\alpha} \\ \underset{\sim}{\alpha} \\ . \\ . \\ \underset{\sim}{\alpha} \end{pmatrix}, \text{where } \underset{\sim}{\alpha} = (\pi_1, \pi_2, ..., \pi_n)$$

with $0 < \pi_j < 1$ and $\sum_{j=1}^{n} \pi_j = 1$

- a ? $\underset{\sim}{P}(t).\underset{\sim}{\pi} = \underset{\sim}{\alpha}$; $t = 0, 1, 2, ...$ and ? $\underset{\sim}{P}(t) = [P_1(t), P_2(t), ..., P_m(t)]$

b $\exists$ constant c and r $(c > 0, 0 < r < 1)$ such that $|P_{i,j}^{(n)} - \pi_j| \leq cr^n, \forall i, j = 1, 2, ..., m$

c $\underset{\sim}{P}\underset{\sim}{\pi} = \underset{\sim}{\pi}\underset{\sim}{P} = \underset{\sim}{\pi}$

Note: The convergence property in b) is known as geometric ergodicity. A chain with his type of property is said to be strongly ergodic. (we are sure that the chain will certainly converge and even faster) Any finite irreducible, time-homogenous Markov Chain $X_m|m \in T$ with transition probability matrix $\underset{\sim}{P}$ has a stationary distribution $\underset{\sim}{V}$ given by

$$\underset{\sim}{V}\underset{\sim}{P} = \underset{\sim}{V}$$

, where $\underset{\sim}{V}$ is a row probability vector which can be found using the equation and the constraint

$$\underset{\sim}{V}\underset{\sim}{I} = 1,$$

with I being a column vector of the same dimension as $\underset{\sim}{V}$ and all entries equal to unity.

$$\underset{\sim}{V} = [v_0, v_1, ..., v_m], \qquad \underset{\sim}{I} = [1, 1, ..., 1]^T_{1\times m}$$

$$\begin{aligned} \underset{\sim}{V}\underset{\sim}{I} &= v_0 + v_1 + ... + v_m \\ &= \sum_{j=1}^{n} v_j \\ &= 1 \end{aligned}$$

A finite irreducible aperiodic Markov Chain $X_m|m \in T$ with a regular transition probability matrix $\underset{\sim}{P}$ has a long run and stationary distribution $\underset{\sim}{\alpha}$ given by $\underset{\sim}{\alpha}\underset{\sim}{P} = \underset{\sim}{\alpha}$ when $\underset{\sim}{\alpha} = (\pi_1, \pi_2, ..., \pi_m)$ assuming

$$\pi_i = lim_{n\to\infty} P X_n = i)$$

using the equation $\underset{\sim}{\alpha}\underset{\sim}{P} = \underset{\sim}{\alpha}$ and $\underset{\sim}{\alpha}I_1 =;$ α can be found.

Note: A stable stochastic matrix is one that has identical rows.

e.g

$$P_{i,j} = \begin{pmatrix} 0.4 & 0.3 & 0.3 \\ 0.4 & 0.3 & 0.3 \\ 0.4 & 0.3 & 0.3 \end{pmatrix}$$

Ever limiting distribution is a stble matrix.

e.g: Suppose a finite hoogeneous Markov Chain has the transition probability matrix:

$$P_{i,j} = \begin{pmatrix} 0 & 2/3 & 1/3 \\ 3/8 & 1/8 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

1. how that the chain is regular and find the long-run distribution.

2. Show that the chain is ergodic and find $lim_{n\to\infty}\underset{\sim}{P}^n$

<u>**Solution**</u>

$\delta = \{0, 1, 2\}$.

The chain is irreducible because all the states $\{0, 1, 2\}$ communicate.

$P_{11}^{(1)} = 1/8 > 0$. The period of state 1 is 1, hence state 1 is aperiodic since periodicity is a class property, then the chain is aperiodic. Since it satisfies the aperiodicity property and all the states communicate, then the chain is regular.

Note: If the chain is regular, then the limiting distribution exists.

$\therefore$, $lim_{n\to\infty}\underset{\sim}{P}^n = \underset{\sim}{\pi}$ exists.

By inference, the chain is ergodic.

Now, let the first row of $\underset{\sim}{\pi}$ be (π_0, π_1, π_2)

$$(\pi_0 \pi_1 \pi_2) \begin{pmatrix} 0 & 2/3 & 1/3 \\ 3/8 & 1/8 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix} = \begin{bmatrix} \pi_0 \\ \pi_1 \\ \pi_2 \end{bmatrix}$$

and $\pi_0 + \pi_1 + \pi_2 = 1$

Solving the system of equations yield $(\pi_0, \pi_1, \pi_2) = (0.3, 0.4, 0.3)$

$\therefore, lim_{n\to\infty} \underset{\sim}{P}^n = \underset{\sim}{\pi} = \begin{pmatrix} 0.3 & 0.4 & 0.3 \\ 0.3 & 0.4 & 0.3 \\ 0.3 & 0.4 & 0.3 \end{pmatrix}$

Assignment: Prove that if a Markov Chain is irreducible and aperiodic and has M states, then the limiting probabilities are given by:

$$\pi_j = \frac{1}{m}; j = 1, 2, ..., m$$

2.5 Classification of States

2.5.1 Recurrence and Transient States

Let $\{X_n\}$ be a Markov Chain with state space $S = \{0, 1, 2, ..., m-1\}$. Define

$$f_{ij}^{(n)} = P(X_n = j, X_r \neq j, r = 1, 2, ..., n-1 | X_0 = i)$$

This means that at time 0, the process was in state i, but in all r steps, the process never got to j until at time n; and

$$f_{ii}^{(*)} = \sum_{n=1}^{\infty} f_{ii}^{(n)} \qquad \text{and} \qquad \mu_{ij} = \sum_{n=1}^{\infty} n f_{ij}^{(n)}$$

$$f_{ii}^{(n)} = P(X_n = i, X_r \neq i, r = 1, 2, ..., n-1 | X_0 = i)$$

1. $f_{ii}^{(*)}$ is the probability that starting from state i, the return to the initial state occurs in a "finite time"

2. $f_{ij}^{(n)}$ is the probability of first passage time when $j = i$, we shall call $f_{ii}^{(n)}$ the recurrence time distribution of state i

3. μ_{ij} is the expected value of the first passage time $\mu_{ii} = \mu_i$ =mean recurrence time

Definition 2.5.1.1. *A state i is said to be recurrent if and only if, starting from i, the eventual return to state i is certain,* $f_{ii}^{(*)} = 1$

Types of recurrence

1. State i is null recurrent if $\mu_i = \infty$

2. State i is recurrent $\mu_i < \infty$

2.5.2 Directed Multigraphs

$M = (I - Q)^{-1}$,$F = MR = f_{ij}$

$\|\sigma_{ij}^2\|^variance = M(2M_D - I) - M_2, i, j \in T$

$$M_D = diag(M) = \begin{bmatrix} \mu_{r+1,r+1} & & & 0 \\ & \ddots & & \\ & & \ddots & \\ & & & \ddots \\ 0 & & & \mu_{m,m} \end{bmatrix}$$

$$M_2 = \begin{bmatrix} \mu_{r+1,r+1}^2 & \mu_{r+1,r+2}^2 & \cdots & \mu_{r+1,m}^2 \\ \vdots & & & \vdots \\ \mu_{m,r+1}^2 & \cdots & \cdots & \mu_{m,m}^2 \end{bmatrix}$$

Chapter 3

Branching Process

Definition 3.0.0.1. *Consider a population of individuals which gives rise to a new population. Assume that the probability of an individual in his lifetime gives rise to r new individuals(offsprings) is P_r for $r = 0, 1, \ldots$ and that individuals are independent rvs. The new population forms a first generation, which in turn reproduces a second generation which in turn produces a third generation etc. For $n = 0, 1, 2 \ldots$ let X_n be the size of the n^{th} generation so that X_0; the $zero_{th}$ generation is initial population then $\{X_n; n = 0, 1, \ldots\}$ is a Markov Chain called a Branching Process. Its states space is $\{0, 1, 2, \ldots\}$.*

Note that 0 is a recurrent state since clearly if $P = P_{ij}$ is TPM, then $P_{00} = 1$.
Also, if $P_0 > 0$, it can be shown that all other states are transient.
Let $f(Z) = \sum\limits_{r=0}^{\infty} PrZ^r, |Z| \leq 1$ be a probability generating fxn (p.g.f)
Note:$f(0) = P_0$ where P_r is the coefficient of Z^r in the expression of f(Z).

$$f(Z) = P_0Z^0 + P_1Z^1 + P_2Z^2 + \ldots$$

$$f(Z) = P_0 + P_1Z + P_2Z^2 + \ldots$$

$$f(0) = 0$$

$$f\prime(Z) = \frac{df}{dt} = P_1 + 2P_2Z + 3P_3Z^2 + \ldots$$

$$f\prime(1) = P_1 + 2P_2 + 3P_3 + \ldots$$

$$= \sum_{r=0}^{\infty} r.P_r$$

= mean number of offsprings of an individual in a single generation

Let $f_n(Z)$ denote the p.g.f of X_n. By considering that

$$X_{n+1} = \sum_{i=1}^{X_n} Z_i$$

where Z_i is the number of offsprings of the i^{th} member of n^th generation.

Thus $Z_i (i \geq 1)$ are independent and identically distributed rvs with distribution $P_r = P(Z_i = r)$, $r = 0, 1, \ldots$ and $\sum_{r=0}^{\infty} P_r = 1$

We assume that $X_0 = 1$.The p.g.f of X_{n+1} is given by

$$f_{n+1}(t) = E(t^{X_{n+1}})$$

$$= E(E[t^{X_{n+1}}|X_n])$$

$$= \sum_{j=0}^{\infty} E(t^{X_{n+1}}|X_n).P(X_n = j)$$

$$= \sum_{j=0}^{\infty} E[t^{\sum_{i=1}^{X_n} Z_i}|X_n].P(X_n = j)$$

$$= \sum_{j=0}^{\infty} E[t^{Z_1+Z_2+\ldots+Z_j}|X_n = j].P(X_n = j)$$

$$= \sum_{j=0}^{\infty} E[\prod_{i=1}^{j} E(t^{Z_i}|X_n = j)].P(X_n = j)$$

Since the Z_i $(i = 1, 2, \ldots, j)$ are independent and identically distributed with p.g.f f(t)

$$\implies f_{n+1}(t) = \sum_{j=0}^{\infty} [f(t)]^j P(X_n = j)$$

$$f_{n+1}(t) = f_n(f(t))$$

$$f_{n+1}(t) = f_n[f(t)]$$

Iterating this relation, we obtain $f_{n+1}(t) = f_n[f(t)]$

$$f_{n+1}(t) = f_{n-1}.f[f(t)]$$

$$= f_{n-1}[f_2(t)]$$

$$= f_{n-2}[f.f_2(t)]$$

$$f_{n+1}(t) = f_{n-2}[f_3(t)]$$

It follows by mathematical induction that $\forall_{k=0,1,\ldots,n}$

$$f_{n+1}(t) = f_{n-k}[f_{k+1}(t)]$$

In particular with $k = n - 1$, we have

$$f_{n+1}(t) = F_1[f_n(t)]$$

$$= F[f_n(t)]$$

Since $X_0 = 1$, $F_1 = f$

Underlying assumptions of a Branching Process

1. $X_0 = 1$

2. None of the probabilities $\{P_r\}$ is equal to 1

3. $0 < P_0 < 1$

4. $P_0 + P_1 < 1$

with these assumptions :the function f(t) is strictly convex on the unit interval of real axis depicted as;

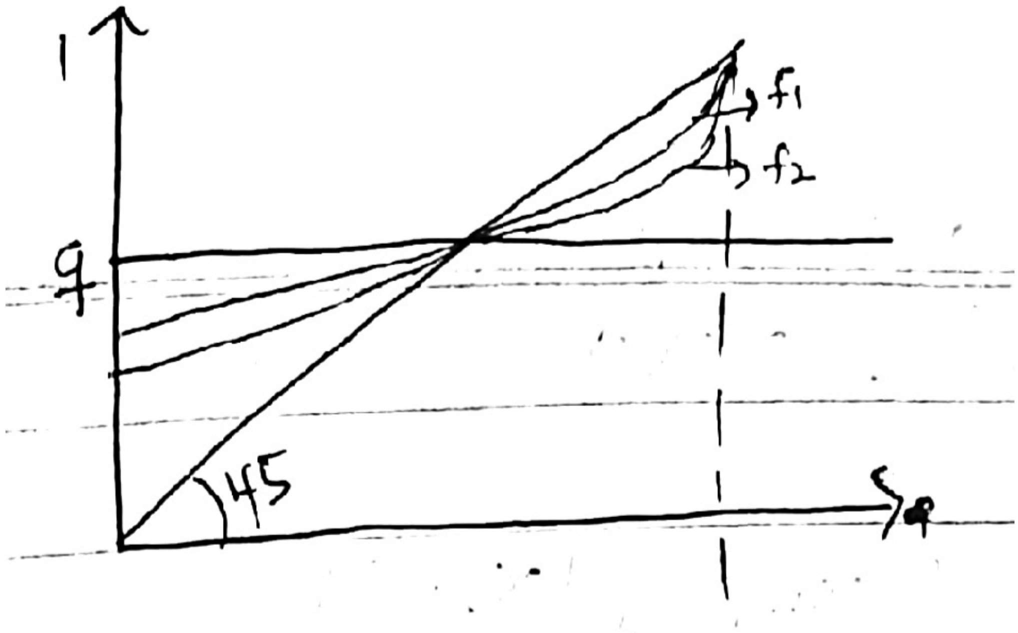

Figure 3.1

Assume the mean and variance of X_1 are finite and let

$$m = E(X_1) = \sum_{r=0}^{\infty} r.P_r$$

$$\sigma^2 = Var(X_1) = E(X_1^2) - [E(X_1)]^2 = \sum_{r=0}^{\infty} r^2 P_r - m^2$$

Now $f_{n+1}(t) = f_n[f(t)] = f[f_n(t)]$

$$f\prime_{n+1}(t) = f\prime(f_n(t))fn\prime(t)........(\star) \text{ chain rule}$$

However, $f\prime n(1) = \sum_{r=1}^{\infty} rP(X_n = r|X_0 = 1)$

if $n = 1$;

$$f\prime(1) = \sum_{r=1}^{\infty} rP(X_1 = r|X_0 = 1)$$

$$= \sum_{r=1}^{\infty} rP(X_1 = r)$$

$$= E(X_1) = m \text{ by independence}$$

$$E(X_n) = m^n \text{ by induction}$$

If we differentiate equation $(\star)$ and put $t = 1$;we find that

$$f\prime\prime_{n+1}(t) = f\prime(f_n(t))f\prime\prime_n(t) + f\prime_n(t)[f\prime\prime(f_n(t))f\prime_n(t)]$$

$$f\prime\prime_{n+1}(1) = f\prime(f_n(1))f\prime\prime_n(1) + f\prime_n(1)[f\prime\prime(f_n(1))f\prime_n(1)]$$

$$f\prime\prime_{n+1}(1) = f\prime(1)f\prime\prime_n(1) + f\prime_n(1)[f\prime\prime(1)f\prime_n(1)]$$

Note $f(t) = E(t^z)$

$$f\prime(t) = E[Z.t^{Z-1}] \qquad f\prime(1) = E(Z)$$

$$f''(t) = E[Z(Z-1)t^{Z-2}] \qquad f''(t) = E(Z^2) - E(Z) \implies E(Z^2) - m$$

$$\implies m[f''_n(1)] + m^n - m^n[\sigma^2 + m^2 - m]$$

$$\implies mf''_n(1) + m^{2n}[\sigma^2 + m^2 - m].......(1)$$

Now:

$$X_{n+1} = \sum_{r=1}^{X_n} Z_r \text{ and } P_{ij} = P(X_{n+1} = j|X_n = i)$$

$$P_{ij} = P\left(\sum_{r=1}^{X_n} Z_r = j|X_n = i\right)$$

$$= P\left(\sum_{r=1}^{X_n} Z_r = j\right)$$

where Z_r is as defined previously. It is clear that X_{n+1} is a r.v and since $Z_r; r = 1, 2, ...$ are iid, we get

$$E[X_{n+1}] = E[X_n].E[Z_r].....(2)$$

$Var[X_{n+1}] = E(X_n)Var(Z_r) + Var(X_n)[E(Z_r)]^2........(3)$

Suppose the population starts with an organism ($X_0 = 1$)

Let $E(Z_r) = \mu$ and $Var(Z_r) = \sigma^2$; $r = 1, 2, \ldots$

From 2:

$$\begin{aligned}
E(X_{n+1}) &= E(X_n)E(Z_r) \\
E(X_n) &= E(X_{n-1})E(Z_r) \\
&= E(X_{n-1})\mu \\
&= [E(X_{n-2})E(Z_r)]\mu \\
&= E(X_{n-2})\mu^2 \\
&= E(X_{n-n+1})\mu^{n-1} \\
&= E(X_{n-n})\mu^n \\
&= E(X_0 = 1)\mu^n \\
E(X_n) &= \mu^n; n \geq 1.......(4)
\end{aligned}$$

Also from 3:

$$\begin{aligned}
Var[X_n] &= E(X_{n-1})Var(Z_r) + Var(X_{n-1})[E(Z_r)]^2 \\
&= \mu^{n-1}\sigma^2 + \mu^2 Var(X_{n-1}) \\
Var(X_n) &= \mu^{n-1}\sigma^2 + \mu^2[\mu^{n-2}\sigma^2 + \mu^2 Var(X_{n-2})] \\
&= \mu^{n-1}\sigma^2 + \mu^n\sigma^2 + \mu^4 Var(X_{n-2})
\end{aligned}$$

$$
\begin{aligned}
Var(X_{n-2}) &= \mu^{n-3}\sigma^2 + \mu^2 Var(X_{n-3}) \\
&= \mu^{n-1}\sigma^2 + \mu^n\sigma^2 + \mu^4[\mu^{n-3}\sigma^2 + \mu^2 Var(X_{n-2}) \\
&= \mu^{n-1}\sigma^2 + \mu^n\sigma^2 + \mu^{n+1}\sigma^2 + \mu^{n-1}\sigma^2 + \underbrace{\mu^6 Var(X_{n-3})} \\
. & \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \to \mu^{2(n-1)} Var(X_{n-(n-1)}) = \mu^{2(n-1)}\sigma^2 \\
Var(X_n) &= \mu^{n-1}\sigma^2 + \mu^n\sigma^2 + \mu^{n+1}\sigma^2 + \mu^{n+2}\sigma^2 + \ldots + \mu^{2n-2}\sigma^2 \\
&= \mu^{n-1}\sigma^2[1 + \mu + \mu^2 + \ldots + \mu^{n-1}] \\
&= \mu^{n-1}\sigma^2 \left[\sum_{r=0}^{n-1} \mu^r\right] \\
&= \mu^{n-1}\sigma^2 \left[\frac{1(1-\mu^n)}{1-\mu}\right], \mu \neq 1 \\
&= n\sigma^2 \text{ if } \mu = 1
\end{aligned}
$$

$$
Var(X_n) = \begin{cases} \mu^{n-1}\sigma^2 \left(\frac{1(1-\mu^n)}{1-\mu}\right) & ; \mu \neq 1 \\ n\sigma^2 & ; \mu = 1 \text{..........(5)} \end{cases}
$$

From (4) and (5), it's clear that the mean and variance of X_n increases or decreases geometrically according as $\mu > 1$ or $\mu < 1$.

Now from the Chebyshev's inequality which states that $E(X_n) = \mu^n < \infty$

and $Var(X_n) < \infty, \exists \epsilon > 0 | P(|X_n - E(X_n)| > \epsilon) \leq \frac{Var(X_n)}{\epsilon^2}$

as $n \to \infty$, when $\mu < 1$

, $\qquad E(X_n) = 0$

$\lim_{n\to\infty} P(|X_n - E(X_n)| > \epsilon) \leq \lim_{n\to\infty} \frac{Var(X_n)}{\epsilon^2}$

$P(|X_n - 0| > \epsilon) \to 0$ because $\lim_{n\to\infty} \mu^{n-1}\sigma^2(\frac{1-\mu^n}{1-\mu})$ but $\lim_{n\to\infty\mu^{n-1}} = 0$

$P(X_n = 0) \to 1$

When $\mu < 1$, then it's certain that the population of size X_n will be extinct as $n \to \infty$

When $\mu \geq$, the probability of extinction of the population is not easy. In this case we may use a generating fxn approach which is also useful in the determination of n-step transition probability of the Branching Process.

Probability of Ultimate Extinction(q)

If the probability of ultimate extinction exist, let it be q.

Clearly $q = \lim_{n\to\infty} P(X_n = 0|X_0 = 1)$

Now the pgf of X_n, $f_n(t) = \sum_{r=0}^{\infty} t^r.P(X_n = r|X_0 = 1)$

$= P(X_n = 0|X_0 = 1) + \sum_{r=1}^{\infty} t^r.P(X_n = r|X_0 = 1)$

$f_n(0) = P(X_n = 0|X_0 = 1)$

$\lim_{n\to\infty} f_n(0) = \lim_{n\to\infty} P(X_n = 0|X_0 = 1) = q$

$\therefore q = \lim_{n\to\infty} f_n(0)$

Generally; $f_{n+1}(t) = f(f_n(t))$

$q = \lim_{n\to\infty} f_{n+1}(0) = f\left[\lim_{n\to\infty} f_n(0)\right]$

$\rightarrow q = f(q)$

Which implies that the probability of ultimate extinction q satisfies

, $q = f(q)$ or $t = f(t)$ if $t = q$

Theorem 3.0.0.2. *The probability of ultimate extinction q exists.*

Proof:

$f_{n+1}(t) = f[f_n(t)]$

Now let $q_n = P(X_n = 0)$

$= f_n(0)$

$q_{n+1} = f_{n+1(0)} = f[f_n(0)]$

$= f(q_n)$

Since $f(t)$ is strictly increasing in t, it is a power series with non-negative co-efficients with $0 < P_0 < 1$.

$$q_1 = f_1(0) - f(0) = P_0 > 0$$

$$q_2 = f_1(q_1) > f(0) = f_1(0) = q_1$$

Assume $q_n > q_{n+1}$

then $q_{n+1} = f(q_n) > f(q_{n-1}) = q_n$

This shows that $q_1, q_2, ..., q_3$ is monotone increasing sequence bounded by 1

Hence $q = \lim_{n \to \infty} q_n$ exists and $0 < q \leq 1$

Theorem 3.0.0.3. *If the mean number of offsprings born to an individual $\mu = E(X_1|X_0 = 1) = f\prime(1)$ is less than or equal to 1, then the probability of ultimate extinction of the population is surely one(1).*

if the mean number is greater than one, then the probability of ultimate extinction is the unique non-negative solution less than 1 of the equation $f = f(t)$ where $f(t) = \sum_{r=0}^{\infty} t^r P_r$

Example: Suppose in a branching process, $P_0 = \frac{1}{2}, P_1 = \frac{1}{4}, P_2 = \frac{1}{4}, P_3 = 0; n \geq 3.$ Determine the probability of the ultimate rxtinction (the prob. that the population will die out.)

solution: $E(X) = \sum_{\forall x} x.P(x) = 0 \times \frac{1}{2} + 1 \times \frac{1}{4} + 2 \times \frac{1}{4} + 3 \times 0$

$E(X) = \frac{3}{4} = \mu < 1$

$\therefore q_n = P(X_n = 0) = 1, n \geq 3$

Example: Suppose in a branching process, $P_0 = \frac{1}{4}, P_1 = \frac{1}{4}, P_2 = \frac{1}{2}, P_n = 0; \forall n \geq 3.$ Show that $q = \frac{1}{2}$

solution: $E(X) = 0 \times \frac{1}{4} + 1 \times \frac{1}{4} + 2 \times \frac{1}{2} = \frac{5}{4}$

$E(X) = \frac{5}{4} = \mu > 1$

$q = f(q)$

$f(q) = \sum_{r=0}^{\infty} q^r.P_r$

$= q^0.P_0 + q^1.P_1 + q^2.P_2$

$= P_0 + q.P_1 + q^2.P_2$

$= \frac{1}{4} + q.\frac{1}{4} + q^2.\frac{1}{2}$

$f(q) = \frac{1}{4} + \frac{q}{4} + \frac{q^2}{4}$

$\frac{1}{4} + \frac{q}{4} + \frac{q^2}{4} = q$

$4q = 1 + q + 2q^2 \implies 2q^2 - 3q + 1 = 0, q = \frac{1}{2}, q = 1$

but we cannot choose $q = 1$ because when $\mu > 1$, prob. of ultimate extinction is not certain.

so $q = \frac{1}{2}$

Chapter 4

The Poisson Process

Let the process $X(t)$ represent the number of times an event occurs in the time $(0, t]$. Define $s < t$

$P_{ij}(s, t) = P(X(t) = j|X(j) = 1$

Further let the event occur under the following postulates.

1. Events occurring in non-overlapping intervals are independent
2. For a sufficiently small Δt, there is a constant λ such that the probability of occurrence of events in $(t, t + \Delta t]$ are given as follows;
   (a) $P_{ii}(t, t + t + \Delta t] = 1 - \lambda\Delta t + 0(\Delta t)$
   (b) $P_{i,i+1}(t, t + \Delta t) = \lambda\Delta t + 0(\Delta t)$
   (c) $\sum\limits_{i+1}^{\infty} P(t, t + \Delta t) = 0(\Delta t)$
   (d) $P_{ij}(t, \Delta t) = 0$; $j < 1$

where $0(\Delta t)$ contains all terms that tends to zero much faster than Δt

ie. $\lim\limits_{\Delta t \to 0} \frac{0(\Delta t)}{\Delta t} = 0$

and $0\Delta t$ represents terms of smaller order than Δt with

$0(\Delta t) + 0(\Delta t) = 0(\Delta t)$ and $c.0(\Delta t) = 0(\Delta t)$, $c \rightarrow$constant

Theorem 4.0.0.1. *Under the above postulates, the number of events occurring in any interval of length t is a Poisson random variable with parameter λt. Thus,*

$$P_n(t) = P(X(t) = n|X(0) = 0 = \frac{e^{\lambda t}.(\lambda t)^n}{n!}, n = 0, 1, ..$$

By the independence assumption of postulate 1 and third option under postulate 2 in conjunction with the Chapman-Kolmogorov Equation we have,

$$P_0(t + \Delta t) = [1 - \lambda \Delta t + 0(\Delta t)].P_0(t) \tag{4.1}$$

$$P_n(t + \Delta t) = P_{n-1}(t)[\lambda \Delta t + 0(\Delta)] + P_n(t)[1 - \lambda \Delta t + 0(\Delta t)] \tag{4.2}$$

$$P_n(t + \Delta t) = \lambda \Delta t P_{n-1}(t) + (1 - \lambda \Delta t)P_n(t) + 0(\Delta t); n > 0$$

From (1):

$$P_0(t + \Delta t) = P_0(t) - \lambda \Delta t.P_0(t) + 0(\Delta t)$$

$$P_0(t + \Delta t) - P_0(t) = -\lambda \Delta t.P_0(t) + 0(\Delta t)$$

$$\frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} = \frac{-\lambda \Delta t.P_0(t)}{\Delta t} + \frac{0(\Delta t)}{\Delta t}$$

$$\frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} = -\lambda.P_0(t) + \frac{0(\Delta t)}{\Delta t} \tag{4.3}$$

From (2):

$$P_n(t + \Delta t) - P_n(t) = \lambda \Delta t.P_{n-1}(t) - \lambda \Delta t.P_n(t) + 0(\Delta t)$$

$$\frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} = \frac{\lambda \Delta t.P_{n-1}(t)}{\Delta t} - \frac{\lambda \Delta t.P_n(t)}{\Delta t} + \frac{0(\Delta t)}{\Delta t}$$

$$\frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} = \lambda.P_{n-1}(t) - \lambda.P_n(t) + \frac{0(\Delta t)}{\Delta t} \tag{4.4}$$

From (3):

$$\lim_{\Delta t \to 0} \frac{P_0(t+\Delta t) - P_0(t)}{\Delta t} = -\lambda.P_0(t) + \lim_{\Delta t \to 0} \frac{0(\Delta t)}{\Delta t}$$

$$P\prime_0(t) = -\lambda P_0(t) \tag{4.5}$$

From (4):

$$\lim_{\Delta t \to 0} \frac{P_n(t+\Delta t) - P_n(t)}{\Delta t} = \lambda P_{n-1}(t) - \lambda.P_n(t) + \lim_{\Delta t \to 0} \frac{0(\Delta t)}{\Delta t}$$

$$P\prime_n(t) = \lambda P_{n-1}(t) - \lambda P_n(t) \tag{4.6}$$

With initial conditions

$$P_n(0) = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{if } n > 0 \end{cases}$$

Eqn(5) and Eqn(6) from a system of different differential equation that can be solved by recursive method as follows:

Multiply both sides of (5) and (6) by $e^{\lambda t}$

$$e^{\lambda t} P\prime_0(t) = -\lambda e^{\lambda t} P_0(t) \tag{4.7}$$

$$e^{\lambda t} P\prime_n(t) = \lambda e^{\lambda t} P_{n-1}(t) - \lambda e^{\lambda t} P_n(t) \tag{4.8}$$

Let $Q_n(t) = e^{\lambda t} P_n(t)$

$$Q\prime_n(t) = \lambda e^{\lambda t} P_n(t) + e^{\lambda t} P\prime_n(t) \tag{4.9}$$

$$Q\prime_0(t) = \lambda e^{\lambda t} P_0(t) + e^{\lambda t} P\prime_0(t) \text{ from (7)}$$

From (8):

$$e^{\lambda t} P\prime_n(t) + \lambda e^{\lambda t} P_n(t) = \lambda e^{\lambda t} P_{n-1}(t)$$

$$Q\prime_n(t) = \lambda e^{\lambda t} P_{n-1}(t) \text{ from (9)}$$

$$Q\prime_n(t) = \lambda Q_{n-1}(t)$$

With the boundary conditions;

$$Q_0(0) = P_0(0) = 1 \text{ and } Q_n(0) = P_n(0) = 1;\ n > 0$$

Now; $\quad Q\prime_0(t) = 0\ ; \qquad \int Q\prime_0(t) = \int 0$

$$Q_0(t) = c$$

Let $t = 0$; $Q_0(0) = 1 \implies c = 1$

$Q_0(t) = 1$

$$Q\prime_1(t) = \lambda Q_0(t)$$

$$Q\prime_1(t) = \lambda$$

$$\int Q\prime_1(t) = \int \lambda$$

$$Q_1(0) = c \therefore c = 0$$

$$Q_1(t) = \lambda t$$

$$Q\prime_2(t) = \lambda Q_1(t) = \lambda^2(t)$$

$$\int Q\prime_2(t) = \int \lambda^2 t$$

$$Q_2(t) = \frac{\lambda^2.t^2}{2} + c$$

$$Q_2(0) = c = 0$$

$$Q_2(t) = \frac{\lambda^2 t^2}{2} = \frac{(\lambda t)^2}{2!}$$

$$Q\prime_3(t) = \lambda Q_2(t) = \frac{\lambda^3 t^2}{2}$$

$$Q_3(t) = \frac{\lambda^3 t^3}{6} + c \quad Q_3(0) = 0 = c$$

$$Q_3(t) = \frac{\lambda^3 t^3}{6} = \frac{(\lambda t)^3}{3!}$$

$$\vdots$$

$$Q_{n-1}(t) = \frac{\lambda^{n-1} t^{n-1}}{(n-1)!}$$

Hence

$$Q\prime_n(t) = \lambda Q_{n-1}(t)$$

$$= \lambda[\frac{\lambda^{n-1}t^{n-1}}{(n-1)!}] = \frac{\lambda^n t^{n-1}}{(n-1)!}$$

$$Q_n(t) = \int \frac{\lambda^n t^{n-1}}{(n-1)!} = \lambda^n . \frac{t^n}{n(n-1)} + c$$

$$Q_n(0) = c = 0$$

$$Q_n(t) = \frac{\lambda^n t^n}{n!}$$

Put

$$Q_n(t) = e^{\lambda t} P_n(t)$$

$$\frac{\lambda^n t^n}{n!} = e^{\lambda t} P_n(t)$$

$$P_n(t) = \frac{e^{\lambda t}(\lambda t)^n}{n!}, n \geq 0$$

Chapter 5

The Renewable Counting Process

WAITING TIME OF THE n^{th} EVENT:

Then T_n is less than or equal to t iff the number of events that have occurred by time t is at least n.

That is if $X(t)$ =the number of events in (0,t], then $P(T_n \leq t) = P(X(t) \geq n)$

$$\sum_{j=n}^{\infty} P(X(t) = j)$$

$$\sum_{j=n}^{\infty} \frac{(\lambda t)^j e^{-\lambda t}}{j!}$$

$$F(t) = \sum_{j=n}^{\infty} \frac{(\lambda t)^j e^{-\lambda t}}{j!}$$

$$f_n(t) = \frac{d}{dt} \sum_{j=n}^{\infty} \frac{(\lambda t)^j e^{-\lambda t}}{j!}$$

$$= \sum_{j=n}^{\infty} \left[\frac{j.(\lambda t)^{j-1}.\lambda.e^{-\lambda t}}{j!} + \frac{(\lambda t)^j (-\lambda e^{-\lambda t})}{j!} \right]$$

$$= \sum_{j=n}^{\infty} \left[\frac{\lambda.e^{-\lambda t}.(\lambda t)^{j-1}}{(j-1)!} - \frac{\lambda e^{-\lambda t} (\lambda t)^j}{j!} \right]$$

$$= \sum_{j=n}^{\infty} \left[\frac{(\lambda t)^{j-1}}{(j-1)!} - \frac{(\lambda t)^j}{j!} \right] \lambda.e^{-\lambda t}$$

$$= \left[\frac{(\lambda t)^{n-1}}{(n-1)!} + \frac{(\lambda t)^n}{n!} + \frac{(\lambda t)^{n+1}}{(n+1)!} + \ldots - \frac{(\lambda t)^n}{n!} - \frac{(\lambda t)^{n+1}}{(n+1)!} - \ldots \right] .\lambda e^{-\lambda t}$$

$$f_n(t) = \frac{(\lambda t)^{n-1}.\lambda e^{-\lambda t}}{(n-1)!} = \frac{\lambda e^{-\lambda t} (\lambda t)^{n-1}}{\Gamma(n)} \sim \Gamma(n, \lambda)$$

Therefor $T_1 \sim exp(\lambda)$ tie of first event.

Note: It is easy to show that if the initial observation of the process is made at s, $s > 0$ at which time

$X(s) = i$

$P_{in}^{s,t} = P(X(t) = n | X_s = i)$

$P_{in}^{s,t} = \frac{e^{-\lambda(t-s)}[\lambda(t-s)^{n-1}]}{(n-1)!}$

This is true as long as λ is time or state dependent.

Theorem: If $X_0 = k$ (and not 1) and q is the probability of ultimate extinction for $X_0 = 1$, the probability of ultimate extinction corresponding to $X_0 = k$ is equal to q^k

Proof:

Since the population will die out iff the families of each of the members of initial generation dies out and since each family is assumed to act independently, the desired probability is q^k.

Question: Suppose that in a discrete branching process, the probability of an individual having k offspring is given by

$$P_k = \frac{e_{-k}\lambda^k}{k!}, k = 0, 1, \ldots$$

However, because the population of interest is heterogenous, λ itself is a random variable distributed distributed according to a gamma distribution

$$g(\lambda) = \begin{cases} \frac{(\frac{q}{p})^{\alpha}.\lambda^{\alpha-1} exp\{\frac{-q}{p}\lambda\}}{\Gamma(\alpha)}; & \lambda \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

where q, p, α are strictly positive constants and $p + q = 1$. Find the p.g.f of the number of offspring of a single individual in the population. Hence find the probability of ultimate extinction.

Solution:

$$X|\lambda \sim P(\lambda)$$

$$f(x,\lambda) = f(x|\lambda).g(\lambda)$$

$$\int_{\forall x} f(x,\lambda)d(\lambda) = \int_{\forall \lambda} f(x|\lambda).g(\lambda)d(\lambda)$$

$$h(x) = \int_{\forall \lambda} f(x|\lambda).g(\lambda)d(\lambda)$$

$$\vdots$$

$$E(t) = \int_{\forall x} t^x.h(x)dx$$

$$g(\lambda) = \frac{(\frac{q}{p})^\alpha.\lambda^{\alpha-1}e^{\{\frac{-q}{p}\lambda\}}}{\Gamma(\alpha)}; \lambda > 0$$

$$\implies \lambda \sim \Gamma(\alpha, \frac{q}{p})$$

$$h(x) = \int_{\forall \lambda} f(x|\lambda).g(\lambda)d\lambda$$

$$= \int_0^\infty \frac{e^{-\lambda}.\lambda^x}{x!}.\frac{(\frac{q}{p})^\alpha.\lambda^{\alpha-1}e^{\{\frac{-q}{p}\lambda\}}}{\Gamma(\alpha)}d\lambda$$

$$= \frac{(\frac{q}{p})^\alpha}{\Gamma(\alpha)x!}\int_0^\infty \lambda^{\alpha+x-1}e^{-\lambda(\frac{q}{p}+1)}d\lambda$$

but

$$\int_0^\infty \lambda^r e^{-\lambda x}d\lambda = \frac{r!}{\lambda^{r+1}}$$

$$\int_0^\infty \lambda^{\alpha+x-1}e^{-\lambda(\frac{q}{p}+1)}d\lambda = \frac{(\alpha+x-1)}{(\frac{q}{p}+1)^{\alpha+x}}$$

$$h(x) = \frac{(\frac{q}{p})^\alpha}{\Gamma(\alpha)x!}.\frac{\Gamma(\alpha+x)}{(\frac{q}{p}+1)^{\alpha+x}}$$

$$= \frac{\Gamma(\alpha+x)}{\Gamma(\alpha)x!}.\frac{(\frac{q}{p})^\alpha}{(\frac{q}{p}+1)^{\alpha+x}}$$

$$= \frac{(\alpha + x - 1)!}{(\alpha - 1)!\, x!} \frac{q^\alpha}{p^\alpha} \div \left(\frac{p+q}{p}\right)^{\alpha + x}$$

$$= \begin{bmatrix} \alpha + x - 1 \\ x \end{bmatrix} \frac{q^\alpha . p^\alpha . p^x}{q^\alpha}$$

$$= \begin{bmatrix} \alpha + x - 1 \\ x \end{bmatrix} \frac{q^\alpha . p^x}{1}$$

$$= \begin{bmatrix} \alpha + x - 1 \\ x \end{bmatrix} .q^\alpha .(1-q)^x ; x = 0, 1, \ldots$$

which is a negative binomial distribution

$$f(t) = E(t^x) = \sum_{\forall x} t^x . h(x)$$

$$\sum_{\forall x}^{\infty} t^x . \begin{pmatrix} \alpha + x - 1 \\ x \end{pmatrix} .q^\alpha .(1-q)^x$$

Identity; $\begin{pmatrix} \alpha + x - 1 \\ x \end{pmatrix} = (-1)^x \begin{pmatrix} -\alpha \\ x \end{pmatrix}$

$$f(t) = \sum_{\forall x}^{\infty} t^x .(-1)^x \begin{pmatrix} -\alpha \\ x \end{pmatrix} .q^\alpha .(1-q)^x$$

$$= \sum_{\forall x}^{\infty} \begin{pmatrix} -\alpha \\ x \end{pmatrix} .q^\alpha .(-tp)^x , p = 1 - q$$

Identity; $(1-t)^{-}\alpha = \sum_{\forall x}^{\infty} \begin{pmatrix} -\alpha \\ x \end{pmatrix} (-t)^x \implies q^\alpha (1 - tp)^{-\alpha}$

$$f(t) = \frac{q^\alpha}{(1-tp)^\alpha} = \left(\frac{q}{(1-tp)}\right)^\alpha$$

To find the prob of extinction, we find the $E(X) = f\prime(1)$.

If it's greater than 1, then we can use $f(t) = t$, and solve for t to get the probability of extinction, otherwise the probability of extinction is 1.

Chapter 6

The Pure(Simple) Birth Process

The assumption of a constant parameter λ in the Poisson Process may not be realistic in physical phenomena such as population growth. A pure general process can be obtained by making the parameter λ dependent on the state of the process.

Consider a process of events occurring under the following postulates. Suppose the event has occurred n times in time $(0, t]$. The the occurrence or non-occurrence of the event during $(t, t + \Delta t]$ for a sufficiently small δt is independent of the time since the last occurrence. Further, the probabilities of events are given as follows:

1. Probability that the event occurs once is $\lambda_n \Delta t + 0(\Delta t))$
2. Probability that the event does not occur is $1 - \lambda_n \Delta t + 0(\Delta t)$
3. Probability that the event occurs more than once is $0(\Delta t)$(negligible)

Let $P_n(t) = P(X(t) = n)$ be the probability that n events occurs in time $(0, t]$. Using the Chapman Kolmogorov equation for transitions in the interval of time $(0, t]$ and $(t, t + \Delta t)$ we have

$$P_n(t + \Delta t) = P_n(t)[1 - \lambda_n \Delta t + 0(\Delta t)] \text{ or } P_{n-1}(t)[\lambda_{n-1} \Delta t + 0(\Delta t)]$$

$$P_n(t + \Delta t) = P_n(t) - \lambda_n \Delta t P_n(t) + \lambda_{n-1} \Delta t P_{n-1}(t) + 0(\Delta t)$$

$$\frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} = \lambda_{n-1} P_{n-1}(t) - \lambda_n P_n(t) + \frac{0(\Delta t)}{\Delta t}$$
$$\lim_{\Delta t \to 0} \frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} = \lambda_{n-1} P_{n-1}(t) - \lambda_n P_n(t) + \lim_{\Delta t \to 0} \frac{0(\Delta t)}{\Delta t}$$

$$P_n{}'(t) = \frac{dP_n(t)}{dt} = \lambda_{n-1} P_{n-1}(t) - \lambda_n P_n(t) \ldots \tag{6.1}$$

Yule Process $\implies \lambda_n = n\lambda$

$$P_n{}'(t) = (n - 1)\lambda P_{n-1}(t) - n\lambda P_n(t) \tag{6.2}$$

Consider the initial population size $X(0) = j$. Then we have the initial conditions

$$P_m(0) = P(X(0) = m) = \begin{cases} 1, & m = j \\ \\ 0, & m \neq j \end{cases}$$

From (6.2), $P_j{}'(t) = (j-1)\lambda P_{j-1}(t) - j\lambda P_j(t)$ but $(j-1)\lambda P_{j-1}(t) = 0$, because if you start from j, you cannot go back to $j-1$ because it's a birth process.

$$\begin{aligned} P_j{}' &= -\lambda j P_j(t) \\ \frac{P_j{}'(t)}{P_j(t)} &= -\lambda j \\ \int \frac{P_j{}'(t)}{P_j(t)} dt &= \int -\lambda j dt \\ \ln P_j(t) &= -\lambda j t + c \\ P_j(t) &= e^{-\lambda j t} \cdot e^c \end{aligned}$$

at $t = 0$, $P_j(0) = 1$, $1 = e^0 \cdot e^c$ so $e^c = 1$

$$\implies P_j(t) = e^{-\lambda j t} \tag{6.3}$$

when $j = n+1$, from 6.2,

$$\begin{aligned}
P_{j+1}'(t) &= \lambda j P_j(t) - \lambda(j+1)P_{j+1}(t) \\
P_{j+1}' &= \lambda j e^{-\lambda j t} - \lambda(j+1)P_{j+1}(t) \\
P_{j+1}'(t) + \lambda(j+1)P_{j+1}(t) &= \lambda j e^{-\lambda j t}
\end{aligned}$$

We use the integrating factor to solve this differential equation;

$$\begin{aligned}
\text{Integrating factor}(I) &= e^{\int \lambda(j+1)dt} \\
I &= e^{\lambda(j+1)t}
\end{aligned}$$

This gives us

$$\begin{aligned}
e^{\lambda(j+1)t}P_{j+1}'(t) + \lambda(j+1)e^{\lambda(j+1)t}P_{j+1}(t) &= \lambda j e^{-\lambda j t} \cdot e^{\lambda(j+1)t} \\
\frac{d}{dt}\left[e^{\lambda(j+1)t}P_{j+1}(t)\right] &= \lambda j e^{\lambda t} dt
\end{aligned}$$

Take integral of both sides

$$\begin{aligned}
e^{\lambda(j+1)t}P_{j+1}(t) &= \int \lambda j e^{\lambda t} dt \\
e^{\lambda(j+1)t}P_{j+1}(t) &= j e^{\lambda t} + c
\end{aligned}$$

At $t = 0$, $P_{j+1}(0) = 0 \implies 0 = j + c \implies c = -j$

$$
\begin{aligned}
e^{\lambda(j+1)t}P_{j+1}(t) &= je^{\lambda t} - j \\
P_{j+1}(t) &= e^{-\lambda(j+1)t}[je^{\lambda t} - j] \\
P_{j+1}(t) &= je^{-\lambda jt} - je^{-\lambda(j+1)t}
\end{aligned}
$$

$$
P_{j+1}(t) = je^{\lambda jt}(1 - e^{-\lambda t}) \tag{6.4}
$$

Thus, preceding in an iterative manner, we can infer that

$$
P_{j+k}(t) = \binom{j+k-1}{j-1} e^{-\lambda jt}(1 - e^{\lambda t})^k, k = 0, 1, \ldots \to \text{negative binomial}
$$

or equivalently,

$$
P_n(t) = \binom{n-1}{j-1} e^{-\lambda jt}(1 - e^{-\lambda t})^{n-j}, n = j, j+1, \ldots
$$

which shows that the population size at time t has a negative binomial distribution in which the probability of success in a single trial is $e^{-\lambda t}$.

Questions

1. Show that for the pure birth process

   (a) $E[X(t)|X(0) = j] = je^{\lambda t}$

   (b) $Var[[X(t)|X(0) = j] = je^{\lambda t}(e^{\lambda t} - 1)$

2. For the pure birth process, find $1(a)$ and $1(b)$ by the method of p.g.f.

3. Derive for the pure birth process the results in $1(a)$ and $1(b)$ using the difference differential equation ${P_n}'(t) = \lambda(n-1)P_{n-1}(t) - \lambda n P_n(t)$

Solution:

1. The pure birth process is negative binomial

   (a) $E[X(t)|X(0) = j] = \frac{j}{e^{-\lambda t}} = je^{\lambda t}$

   (b) $Var[[X(t)|X(0) = j] = \frac{j(1-e^{-\lambda t})}{(e^{-\lambda t})^2} = j - je^{-\lambda t} \cdot (e^{\lambda t})^2 = je^{\lambda t}(e^{\lambda t} - 1)$

Chapter 7

The Pure(Simple) Death Process

Suppose an initial size, say $i > 0$ individuals die at a certain rate, eventually reducing the size to zero. When the population size is n, let u_n be the death rate defined as follows: in an interval $(t, t + \Delta t]$, the probability that one death occurs in the interval is $u_n \Delta t + 0(\Delta t)$, while the probability that no death occurs is $1 - u_n \Delta t + 0(\Delta t)$, and all other probabilities are negligible, $0(\Delta t)$.Also assume that the occurrence of death in this interval $(t, t + \Delta t]$ is independent of time since the last death. Let $P_n(t) = P[X(t) = n]$ be the probability that there are n individuals in the population within time interval $(0, t]$. By Chapman-Kolmogorov equations for transitions in the intervals of time $(0, t]$ and $(t, t + \Delta t]$, we have:

$$
\begin{aligned}
P_n(t + \Delta t) &= P_n(t)[1 - \mu \Delta t + 0(\Delta t)] + P_{n+1}[\mu_{n+1} \Delta t + 0(\Delta t)] \\
P_n(t + \Delta t) &= P_n(t) - \mu_n \Delta t P_n(t) + \mu_{n+1} \Delta t P_{n+1}(t) + 0(\Delta t) \\
\frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} &= \mu_{n+1} P_{n+1}(t) - \mu_n P_n(t) + \frac{0(\Delta t)}{\Delta t} \\
\lim_{\Delta t \to 0} \frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} &= \mu_{n+1} P_{n+1}(t) - \mu_n P_n(t) + \lim_{\Delta t \to 0} \frac{0(\Delta t)}{\Delta t}
\end{aligned}
$$

$$P_n{}'(t) = \mu_{n+1}P_{n+1}(t) - \mu_n P_n(t) \tag{7.1}$$

From the Yule's process, $\mu_n = n\mu$

$$P_n{}'(t) = P_{n+1}(t) - \mu_n P_n(t) \tag{7.2}$$

Consider the initial conditions

$$P_m(0) = \begin{cases} 1, & m = j \\ 0, & m \neq j \end{cases}$$

From 7.2, $P_j{}'(t) = \mu(j+1)P_{j+1}(t) - \mu j P_j(t)$.

$\mu(j+1)P_{j+1}(t)$ is 0 because you can't get $j+1$ from j if you're in a death process.

$$\begin{aligned}
P_j{}'(t) &= -\mu j P_j(t) \\
\frac{P_j{}'(t)}{P_j(t)} &= -\mu j \\
\int \frac{P_j{}'(t)}{P_j(t)} dt &= \int -\mu j dt \\
\ln P_j(t) &= -\mu j t + c \\
P_j(t) &= e^{-\mu j t} \cdot e^c
\end{aligned}$$

At $t = 0$, $P_j(0) = 1$ so $e^c = 1$ which implies $P_j(t) = e^{-\mu j t} = (e^{-\mu t})^j$.

NOTE: After the first person dies, you go to $j-1$ not $j+1$. When $n = j-1$,

$$P'_{j-1}(t) = \mu j P_j(t) - \mu(j-1)P_{j-1}(t)$$

$$P'_{j-1}(t) + \mu(j-1)P_{j-1}(t) = \mu j P_j(t)$$

$$I = e^{\int \mu(j-1)dt} = e^{\mu(j-1)t}$$

$$\begin{aligned}
e^{\mu(j-1)t}P_{j-1}(t) + \mu(j-1)e^{\mu(j-1)t}P_{j-1}(t) &= \mu j e^{\mu(j-1)t}P_j(t)\\
e^{\mu(j-1)t}P_{j-1}(t) &= \int \mu j e^{\mu(j-1)t}P_j(t)dt\\
e^{\mu(j-1)t}P_{j-1}(t) &= \int \mu j e^{\mu(j-1)t}e^{-\mu tj}dt\\
e^{\mu(j-1)t}P_{j-1}(t) &= \int \mu j e^{-\mu t}dt\\
&= -je^{-\mu t} + c
\end{aligned}$$

At $t = 0$, $P_{j-1}(0) = 0$ so $c = j$.

$$\begin{aligned}
e^{\mu(j-1)t}P_{j-1}(t) &= -je^{-\mu t} + j\\
P_{j-1}(t) &= e^{-\mu(j-1)t}(-je^{-\mu t} + j)\\
P_{j-1}(t) &= -je^{-\mu jt} + je^{-\mu(j-1)t}\\
P_{j-1}(t) &= je^{-\mu(j-1)t}(1 - e^{-\mu t})
\end{aligned}$$

Hence, preceding in an iterative manner, we infer that

$$P_n(t) = \binom{j}{n}(e^{-\mu t})^n(1 - e^{-\mu t})^{j-n}, n = 0, 1, \ldots, j \sim bin(j, e^{-\mu t})$$

This shows that the population size is binomially distributed with mean and variance given by

1. $E[X(t)|X(0) = j] = je^{-\mu t}$
2. $Var[X(t)|X(0) = j] = je^{-\mu t}(1 - e^{-\mu t})$

Chapter 8

Birth And Death Process

Birth is an event which signifies an increase in the population. Death is an event which signifies a decrease in population. Suppose these two types of events occur under the following postulates:

1. Birth: If the population size n (20) at time t during the following infinitesimal interval (t,t+Δt), the probability that birth will occur is $\lambda_n\Delta$t + $0(\lambda t)$

   $\lim\limits_{\Delta t \to 0} \frac{0(\Delta t)}{\Delta t} = 0$ hence 1 - $\lambda_n \Delta t + 0(\Delta t)$

   Births occurring in (t, t+ Δt] are independent of time since the last occurrence.

2. Death: If the population size n $>$ 0 at time t during the following infinitesimal interval (t,t+Δt) and 1 - $\mu_n \Delta t$ is no death occurring in the interval (t,t+Δt). Death occurring in (t,t+Δt) are independent of time since the last occurrence

3. When the population size is 0 at time t, the probability is 0 that death occurring during (t,t+Δt).

   Note: More than one change (birth or death) takes palce in $0(\Delta t)$

4. For the same population size, births and deaths occur independent of each other.

Let X(t) be the population size at t. Define

$$P_{i,n}^{s,t} = P(X(t) = n | X(s) = i)$$

This process is time homogenous and therefore we shall use the definition given by

$$P_n^t = P(X(t) = n | X(0) = i)$$

As a consequence of postulates 1-4, we may write

$$P_{n,n-1}^{t,t+\Delta t} = [\mu_n \Delta t + 0(\Delta t)][1 - \lambda_n \Delta t + 0(\Delta t)]$$

$$= \mu_n \Delta t + \mu_n \Delta t \lambda_n \Delta t + 0(\Delta t)$$

$$= \mu_n \Delta t \qquad ... (1)$$

$$P_{n,n}^{t,t+\Delta t} = [1 - \mu_n \Delta t + 0(\Delta t)][1 - \lambda_n \Delta t + 0(\Delta t)]$$

$$1 - \lambda_n \Delta t - \mu_n \Delta t + 0(\Delta t) \qquad ... (2)$$

$$P_{n,n+1}^{t,t+\Delta t} = [\lambda_n \Delta t + 0(\Delta t)][1 - \mu_n \Delta t + 0(\Delta t)]$$

$$= \lambda_n \Delta t + 0(\Delta t) \qquad ... (3)$$

$$\sum_{j \neq n-1, n+1} P_{nj}(t, t + \Delta t) = 0(\Delta t) \qquad ... (4)$$

Any other steps that do not reduce the population by 1 or increase it by 1 moves towards 0. eg n-2 = 0(Δt), n+2 = 0(Δt)

For transitions occurring in non-overlapping intervals (0,t] and (t,t+Δ t] based on equations (1) through (4), then the Chapman-Kolmogorov equations will take the form

$$P_n(t+\Delta t) = [1-\lambda_n\Delta t - \mu_n\Delta t + 0(\Delta t)]P_n(t) + [\lambda_n\Delta t + 0(\Delta t)]P_{n-1}(t) + [\mu_n\Delta t + 0(\Delta t)P_{n+1}(t)]$$

$$P_n(t+\Delta t) = (1-\lambda_n\Delta t - \mu_n\Delta t)P_n(t) + [\lambda_{n-1}\Delta t]P_{n-1}(t) + [\mu_{n+1}\Delta t]P_{n+1}(t) + 0(\Delta t)$$

$$P_n(t+\Delta t) = P_n(t) - (\lambda_n\Delta t + \mu_n\Delta t)P_n(t) + \lambda_{n-1}\Delta t P_{n-1}(t) + \mu_{n+1}\Delta t P_{n+1} + 0(\Delta t)$$

$$\lim_{\Delta t\to 0}\frac{P_n(t+\Delta t)-P_n(t)}{\Delta t} = -(\lambda_n+\mu_n)P_n(t) + \lambda_{n-1}P_{n-1}(t) + \mu_{n+1}P_{n+1}(t) + \lim_{\Delta t\to 0}\frac{0(\Delta t)}{\Delta t}$$

$$P_n'(t) = -(\lambda_n+\mu_n)P_n(t) + \lambda_{n-1}P_{n-1}(t) + \mu_{n+1}P_{n+1}(t), n = 0,1,2...$$

With initial conditions, $P_n(0) = \begin{cases} 1 & \text{if n=i and } P_{n-1}(t) = 0 \\ \\ 0 & \text{if n } \neq i \end{cases}$ If $\lambda_n = n\lambda$ and $\mu_n = n\mu$

Then the simple linear birth and death process becomes

$$P_n'(t) = -n(\lambda+\mu)P_n(t) + (n-1)\lambda P_{n-1}(t) + (n+1)\mu P_{n+1}(t)... \tag{6}$$

$$E[X(t) = n|X(0) = i] = E(X(t)) = \sum_{n=0}^{\infty} nP_n(t)$$

$$\frac{d}{dt}E[X(t)] = \sum_{n=0}^{\infty} n\frac{d}{dt}P_n(t) = \sum_{n=0}^{\infty} nP_n'(t)$$

$$\sum_{n=0}^{\infty} n[-n(\lambda+\mu)P_n(t) + (n-1)\lambda P_{n-1}(t) + (n+1)\mu P_{n+1}(t)]$$

$$-\sum_{n=0}^{\infty} n^2(\lambda+\mu)P_n(t) + \lambda\sum_{n=0}^{\infty} n(n-1)P_{n-1}(t) + \mu\sum_{n=0}^{\infty} n(n+1)P_{n+1}(t)$$

let n-1=n and n+1=n

$$= -(\lambda+\mu)\sum_{n=0}^{\infty} n^2P_n(t) + \lambda\sum_{n=0}^{\infty}(n+1)nP_n(t) + \mu\sum_{n=0}^{\infty}(n-1)nP_n(t)$$

$$\frac{d}{dt}E[X(t)] = (\lambda-\mu)E[X(t)]$$

$(\lambda - \mu)$ is the intrinsic growth rate

$$\frac{dE[X(t)]}{E[X(t)]} = (\lambda-\mu)dt$$

take integral of both sides

$$\int\frac{dE[X(t)]}{E[X(t)]} = \int(\lambda-\mu)dt$$

$$=> \ln E[X(t)] = (\lambda-\mu)\text{t + c}$$

$$E[X(t)] = e^{(\lambda-\mu)t}e^c$$

$$X(0) = i \quad i = e^{(\lambda-\mu)(0)}.e^c => e^c = i$$

$$E[X(t)|X(0) = i] = ie^{(\lambda-\mu)t}$$

Recall equation 6

$$P_0(t) = \epsilon_t = \mu\frac{[1 - e^{-(\lambda-\mu)t}]}{[\lambda - \mu e^{(\lambda-\mu)}]}$$

$$P_n(t) = (1 - \epsilon_t)(1 - n_t)n_t^{n-1}$$

$$\text{where } n_t = \left(\frac{\lambda}{\mu}\right)\epsilon_t = \frac{\lambda[1 - e^{-(\lambda-\mu)t}]}{\lambda - e^{-(\lambda-\mu)t}}$$

Clearly $P_n(t)$ has a geometric distribution modified with the initial term. $\lim_{t\to\infty} P_0(t)$=Probability of ultimate extinction of the population.

$$\text{let } \rho = \frac{x}{\mu}; \lim_{t\to\infty} \epsilon_t = \mu\left[\frac{1-0}{\lambda-0}\right] = \frac{\mu}{\lambda} = \rho^{-1}$$

$$\text{and } \lim_{t\to\infty} \epsilon_t = \lim_{t\to\infty} \frac{e^{(\mu-\lambda)t}}{e^{(\mu-\lambda)t}}\left[\frac{\mu e^{-(\mu-\lambda)t} - \mu}{\lambda e^{-(\mu-\lambda)t} - \mu}\right]; \rho \le 1$$

$$\frac{\lambda}{\mu} => \lambda \le \mu$$

$$=> \lim_{t\to\infty} P_0(t) = \begin{cases} 1 & \text{if } \rho \le 1 \text{ or } \lambda \le \mu \\ \rho^{-1} & \text{if } \rho > \text{ or } \lambda > \mu \end{cases}$$

Meaning, ultimate extinction is certain if death rate is larger than birth rate Meaning ultimate extinction is certain if death rate is larger than birth rate. **Question:** The p.g.f.of $X(t)$, the number of individuals in the population at time t of the simple linear pure birth and death process with $X(0) = 1$ is

$$\phi(z,t) = \begin{cases} \frac{\mu(1-\alpha)-(\lambda-\mu\alpha)z}{\mu-\lambda\alpha-\lambda(1-\alpha)z}, & \text{if } \lambda \neq \mu \\ \frac{1-(\lambda t-1)(z-1)}{1-\lambda t(z-1)}, & \text{if } \lambda = \mu \end{cases}$$

where $\alpha = e^{(\lambda-\mu)t}$. Use the above p.g.g. to derive an expression for

1. $P_0(t)$ when $\lambda \neq \mu$.
2. $P_0(t)$ when $\lambda = \mu$

3. $P_n(t), n \geq 1$ when $\lambda \neq \mu$

Hence show that

$$\lim_{t \to \infty} P_0(t) = \begin{cases} 1, & \text{if } \lambda \leq \mu \\ \\ \frac{\mu}{\lambda}, & \lambda > \mu \end{cases}$$

and comment on the results with special reference to the intrinsic growth rate$(\lambda - \mu)$.

Comments: The generalized birth and death process is a continuous Markov process. It is said to be time-homogeneous because the parameters λ_n and μ_n do not depend on time, but they depend only on the size of the population($n = 0, 1, 2, \dots$). If $\lambda_0 = 0$, then state 0 becomes an absorbing state(recurrent). All states are transient, that is, the process is not irreducible to this case. In the case where $\lambda_n = n\lambda$, and $\mu_n = n\mu(n \geq 1)$ the simple birth and death or simple linear growth process, where $\lambda_0 = 0$ and for $X(0) = 1$, then the ultimate extinction probability

$$\lim_{t\to\infty} P_0(t) = \begin{cases} 1, & \text{if } \lambda \leq \mu \\ \\ \frac{\mu}{\lambda}, & \lambda > \mu \end{cases}$$

Theorem 8.0.0.1. *Where $X(0) = i$, $\lambda_n = n\lambda$ and $\mu_n = n\mu(n \geq 1)$, the corresponding absorption(ultimate extinction) probabilities are*

$$\lim_{t\to\infty} P_0(t) = \begin{cases} 1, & \textit{if } \lambda \leq \mu \\ \\ (\frac{\mu}{\lambda})^i, & \lambda > \mu,\ i \textit{ is the initial size of the population} \end{cases}$$

The preceding theorem can be proved by noting that since we have assumed independence and no interaction among the members, we may view the population as the sum of i independent simple birth and death processes each beginning with a single member thus if $\phi(z, t)$ is the p.g.f. corresponding to $X(0) = 1$, then the p.g.f. corresponding to $X(0) = i$ is

$$\psi(z,t) = \begin{cases} \left[\frac{\mu(1-\alpha)-(\lambda-\mu\alpha)z}{\mu-\lambda\alpha-\lambda(1-\alpha)z}\right]^i, & \text{if } \mu = \lambda \\ \left[\frac{1-(\lambda t-1)(z-1)}{1-\lambda t(z-1)}\right]^i, & \text{if } \mu \neq \lambda \end{cases}$$

where $\alpha = e^{(\lambda-\mu)t}$.

Definition 8.0.0.2. *A birth and death process is called a linear growth process with immigration if $\lambda_n = n\lambda + \epsilon$ and $\mu_n = n\mu$ with $\lambda, \mu, \epsilon > 0$, such processes occur naturally in the study of the biological reproduction and population growth.*

If the state n describes the current population size, then the average instantaneous rate of growth is $\lambda_n + \epsilon$. The probability of the state of the process decreasing by one in a small time interval of length Δt is $\mu_n(\Delta t) + 0(\Delta t)$. The factor λ_n represents the natural growth of the population owing to its current size while the second factor ϵ may be interpreted as the infinitesimal rate of increase of the population due to an external source such as immigration.

Note: This process is irreducible and therefore there is no absorbing state. The probability of ultimate extinction is zero for $\lambda_0 = \epsilon > 0$. Consider a population subject to death, birth, emigration and immigration under the following assumptions:

1. In a small time interval of length Δt, the probability for a given individual to
   (a) die or emigrate is $\mu\Delta t$
   (b) give birth to a new individual $\lambda\Delta t$
2. Immigrants are subject to death, emigration and give birth in the same manner as existing members.
3. In a small time interval of length Δt, the probability of the population being
   (a) increased by one immigrant is $\epsilon\Delta t + 0(\Delta t)$.
   (b) increased or decreased by more than one is $0(\Delta t)$.

Question: Based on the above assumptions, if $X(t)$ is the total population size at time t and $P_n(t) = P[X(t) = n]$; $n = 0, 1, 2, \ldots$,prove that

$$\frac{d}{dt}P_n(t) = [\lambda(n-1) + \epsilon]P_{n-1}(t) + \mu(n+1)P_{n+1}(t) - [n(\lambda + \mu) + \epsilon]P_n(t)$$

.

Deduce that if $P_n(0) = \begin{cases} 1, & \text{if } n = i \\ 0, & \text{if } n \neq i \end{cases}$ then

$$E[X(t) = n|X(0) = i] = \begin{cases} \frac{\epsilon}{\lambda-\mu}\left[e^{(\lambda-\mu)t} - 1\right] + ie^{(\lambda-\mu)t}, & \text{if } \lambda \neq \mu \\ \epsilon t + i, & \text{if } \lambda = \mu \end{cases}$$

Solution:

Let

$$\begin{aligned} P_{in}^{(s,t)} &= P[X(t) = n|X(s) = i] \\ P_n(t) &= P_{in}(0,t) = P[X(t) = n|X(0) = i] \\ P_{n,n-1}(t, t+\Delta t) &= [\mu n\Delta t + 0(\Delta t)][1 - \lambda n\Delta t + 0(\Delta t)][1 - \epsilon\Delta t + 0(\Delta t)] \\ &= \mu n\Delta t + 0(\Delta t) \end{aligned}$$

$$\begin{aligned} P_{n,n}(t, t+\Delta t) &= [1 - \mu n\Delta t + 0(\Delta t)][1 - \lambda n\Delta t + 0(\Delta t)][1 - \epsilon\Delta t + 0(\Delta t)] + [\lambda n\Delta t + 0(\Delta t)][\mu n\Delta t + 0(\Delta t)][1 - \epsilon\Delta \\ &= 1 - [n(\lambda + \mu) + \epsilon]\Delta t + 0(\Delta t) \end{aligned}$$

$$\begin{aligned} P_{n,n}(t, t+\Delta t) &= [\mu n\Delta t + 0(\Delta t)][\lambda n\Delta t + 0(\Delta t)][\epsilon\Delta t + 0(\Delta t)] + [1 - \mu n\Delta t + 0(\Delta t)][\lambda n\Delta t + 0(\Delta t)][1 - \epsilon\Delta t + 0(\Delta \\ &= (\lambda n + \epsilon)\Delta t + 0(\Delta t) \end{aligned}$$

$$\sum_{j \neq n-1, n, n+1} P_{nj}(t, t+\Delta t) = 0(\Delta t) \tag{8.1}$$

from Chapman-Kolmogorov equation for Markov process with discrete state space.

But

$$P_{n-1,n}(t, t+\Delta t) = [\lambda n \Delta t + 0(\Delta t)][1 - \epsilon \Delta t + 0(\Delta t)][1 - \mu n \Delta t + 0(\Delta t)] + [1 - \lambda n \Delta t + 0(\Delta t)][1 - \mu n \Delta t + 0(\Delta t)]$$

$$= [\lambda(n-1) + \epsilon]\Delta t + 0(\Delta t)$$

This implies

$$P_n(t+\Delta t) = [1 - [n(\lambda+\mu) + \epsilon]\Delta t + 0(\Delta t)] \cdot P_n(t) + [[\lambda(n-1) + \epsilon]\Delta t + 0(\Delta t)] \cdot P_{n-1}(t) + [\mu(n+1)\Delta t + 0(\Delta t)]$$

$$= P_n(t) - [n(\lambda+\mu) + \epsilon]\Delta t \cdot P_n(t) + [\lambda(n-1) + \epsilon]\Delta t P_{n-1}(t) + \mu(n+1)\Delta t P_{n+1}(t) + 0(\Delta t)$$

$$\lim_{\Delta t \to 0} \frac{P(t+\Delta t) - P_n(t)}{\Delta t} = -[n(\lambda+\mu) + \epsilon]P_n(t) + [\lambda(n-1) + \epsilon]P_{n-1}(t) + \mu(n+1)P_{n+1}(t) + \lim_{\Delta t \to 0} \frac{0(\Delta t)}{\Delta t}$$

$$\frac{d}{dt}P_n(t) = -[n(\lambda+\mu) + \epsilon]P_n(t) + [\lambda(n-1) + \epsilon]P_{n-1}(t) + \mu(n+1)P_{n+1}(t)$$

$$E[X(t)|X(0) = i] = \sum_{n=0}^{\infty} n \cdot P_n(t)$$

$$\frac{d}{dt}E[X(t)] = \sum_{n=0}^{\infty} n \cdot \frac{d}{dt}P_n(t)$$

$$= \sum_{n=0}^{\infty} n[[-n(\lambda+\mu) + \epsilon]P_n(t) + [\lambda(n-1) + \epsilon]P_{n-1}(t) + \mu(n+1)P_{n+1}(t)]$$

$$= -(\lambda+\mu)\sum_{n=0}^{\infty} n^2 P_n(t) + \epsilon \sum_{n=0}^{\infty} nP_n(t) + \lambda \sum_{n=0}^{\infty} n(n-1)P_{n-1}(t) + \epsilon \sum_{n=0}^{\infty} nP_{n-1}(t) + \mu \sum_{n=0}^{\infty} n(n+1)P$$

Theorem 8.0.0.3. *If a Markov Process is irreducible, then the limiting distribution* $\lim_{n\to\infty} P_n$ *exists and is independent of the initial conditions of the process. The limits(*$P_n, n \in S$*, the state space) are such that they either vanish identically, i.e.*$P_n = 0 \ \forall n \in S$ *or are all positive and form a probability distribution or vector, i.e.*$P_n > 0 \ \forall n \in S$ *and* $\sum_{n\in S} P_n = 1$.

Note: In this case $P_n\prime(t) \to 0$ as $t \to \infty$, thus for the birth and death process with immigration, whether or not a steady rate distribution exists if $X(0) = i$ and $\mu > \lambda$, $P_n\prime(t) \to 0$ as $t \to \infty$ and

$$[\lambda(n-1) + \epsilon]P_{n-1}(t) + (n+1)\mu P_{n+1}(t) - [n(\mu + \lambda) + \epsilon]P_n(t) = 0$$